

AI Attendance System Using Face Recognition

TESTING

Testing Approach

Testing is an important step in the development of the AI Attendance System using Face Recognition. The purpose of testing is to ensure that the system works correctly, accurately, and reliably in different conditions. The system is tested to verify that each module performs its function properly and that the complete system meets the requirements.

The testing process follows a step-by-step approach. First, individual modules such as image capture, face detection, face recognition, and attendance storage are tested separately. After confirming that each module works correctly, integration testing is performed to check whether all modules work together smoothly.

Different types of testing methods are used:

Unit Testing:

Each module is tested individually. For example, the face detection module is tested to ensure that it correctly identifies faces from images.

Integration Testing:

All modules are combined and tested together. The system is checked to ensure that captured images are correctly processed and attendance is recorded properly.

System Testing:

The complete system is tested as a whole to ensure that it meets the project requirements.

Performance Testing:

The system is tested with multiple users and different lighting conditions to check speed and accuracy.

User Testing:

The system is tested by users to ensure that it is easy to use and understand.

The testing approach helps in identifying errors and improving system performance.

Sample Test Plan

The test plan describes different test cases used to verify the system functionality.

Test Case 1: Camera Functionality

Objective:

Check whether the webcam captures images properly.

Input:

Start camera module.

Expected Result:

Camera should open and display video.

Status:

Pass

Test Case 2: Face Detection

Objective:

Verify that the system detects a human face.

Input:

Show face in front of camera.

Expected Result:

Face should be detected and highlighted.

Status:

Pass

Test Case 3: Face Recognition

Objective:

Check whether the system correctly identifies registered users.

Input:

Registered user stands in front of camera.

Expected Result:

System should display correct user name.

Status:

Pass

Test Case 4: Unknown Person Detection

Objective:

Check how the system responds to unregistered users.

Input:

Unregistered person stands in front of camera.

Expected Result:
System should display "Unknown".

Status:
Pass

Test Case 5: Attendance Recording

Objective:
Verify attendance storage.

Input:
Recognized user detected.

Expected Result:
Attendance should be recorded with date and time.

Status:
Pass

Test Case 6: Duplicate Attendance Prevention

Objective:
Ensure attendance is not marked multiple times.

Input:
Same user appears again.

Expected Result:
Attendance should not be recorded again for same session.

Status:
Pass

Sample Test Results

The system is tested with multiple users under different lighting conditions. The results show that the system detects and recognizes faces accurately when the face is clearly visible. The attendance is recorded correctly with date and time. The system performs faster when good lighting is available.

Accuracy:
The system provides high accuracy when trained with sufficient images.

Speed:
Face detection and recognition occur within a few seconds.

Reliability:

System works consistently when camera is stable.

Limitations observed during testing:

- Poor lighting may reduce accuracy.
- Face should be clearly visible to camera.
- Performance depends on image quality.

Necessary improvements such as increasing training images can improve accuracy further.

Conclusion

Testing confirms that the AI Attendance System works correctly and meets the project requirements. The system successfully detects and recognizes faces and records attendance automatically. The testing process helps ensure accuracy, reliability, and proper performance. The results show that the system is suitable for practical use in educational institutions.